

05
CVP

Class - B2

7/12/25

1st Paper

Chapter - 6

Long-term Financing

Bond :- Debt instrument (Debt capital) -
Long term liability

Obligation :- { 1. Periodic interest
2. Principal repayment at maturity }

* Treasury Bond :- [Risk Free Return]

{ 1. Issued by govt. [default]
2. Return is guaranteed. }

* Call Bond

$$\underline{CIR} > \underline{MIR}$$

* Put Bond

$$\underline{CIR} < \underline{MIR}$$

CW

Class - 3B

27/12/25

1st Paper

Chapter - 6

Bond $\rightarrow$ asset security - secured bond

Debenture $\rightarrow$ no $\rightarrow$ unsecured bond

Optimal Capital Structure: The combination of equity and debt capital in which a company's

value will be ~~maximized~~ maximized and cost of capital will be lowest.

$$\frac{CIR}{WACC} < 1$$

* Govt. bond

$$\frac{CIR}{WACC} > 1$$

CWO

1st Paper / Class - B4

11.12.23

Chapter - 6Valuation of Bond* Coupon Bond

$$V_B = \underbrace{\frac{I}{m} \times \left\{ \frac{1 - \frac{1}{(1 + \frac{k_d}{m})^{n \times m}}}{\frac{k_d}{m}} \right\}}_{\text{Periodic Interest}} + \underbrace{\frac{MV}{(1 + \frac{k_d}{m})^{n \times m}}}_{\text{Principal amount}}$$

Periodic Interest

Here, $50 \times 100 = 5000$

$I = \text{Coupon interest} = \text{intake} =$

$MV = \text{Maturity value}$

$m = \text{no. of compounding.}$

* Apex Company issued 10 years 10%.

Bond at Tk. 1,000. $k_d = 12\%$. Find the value of the bond.

Solⁿ: Given,

$$FV = \text{Tk. } 1,000$$

$$CR = 10\%$$

$$I = 1,000 \times 10\%$$

$$= 100$$

$$k_d = 12\% \quad , \quad MV = \text{Tk. } 1,000$$

$$\therefore V_B = \frac{I}{k_d} \times \left\{ \frac{1 - \frac{1}{(1+k_d)^n}}{k_d} \right\} + \frac{MV}{(1+k_d)^n}$$

$$= 100 \times \left\{ \frac{1 - \frac{1}{(1+0.12)^{10}}}{0.12} \right\} + \frac{1000}{(1+0.12)^{10}}$$

$$= 100 \times 5.6502 + 321.0732$$

$$= 565.02 + 321.0732$$

$$= \text{Rs. } 886.09$$

(Ans.)

★

$CIR < k_d \rightarrow V_B < \text{Face value} \rightarrow \text{discount}$

$CIR = k_d \rightarrow V_B = \text{Face value} \rightarrow \text{par value}$

$CIR > k_d \rightarrow V_B > \text{Face value} \rightarrow \text{premium}$

$$500,000 = V_B$$

$$CIR = 10\%$$

$$500,000 = C$$

$$500 =$$

$$500,000 = V_B, CIR = 10\%$$

Q11W1

18/12/25

Ch. 6

1 we know,

$$V_B = I \times \left[\frac{1 - \frac{1}{(1+kd)^n}}{kd} \right] + \frac{MV}{(1+kd)^n}$$

$$= 100 \times \left[\frac{1 - \frac{1}{(1+0.1)^{10}}}{0.1} \right] + \frac{1,000}{(1+0.1)^{10}}$$

$$= 100 \times 6.1446 + 385.5432$$

$$= \text{Tk. } 1,000$$

$$\therefore CIR = kd, V_B = FV.$$

Given,

$$FV = \text{Tk. } 1,000$$

$$kd = 0.1$$

$$CIR = 0.1$$

$$MV = 1000 \text{ Tk.}$$

$$n = 10$$

$$I = FV \times CIR$$

$$= 100$$

2 we know,

$$V_B = I \times \left[\frac{1 - \frac{1}{(1+kd)^n}}{kd} \right] + \frac{MV}{(1+kd)^n}$$

$$= 100 \times \left[\frac{1 - \frac{1}{(1+0.08)^{10}}}{0.08} \right] + \frac{1,000}{(1+0.08)^{10}}$$

$$= 100 \times 6.71 + 463.1034$$

$$= \text{Tk. } 1,134.1034$$

$$\therefore CIR > kd, V_B > FV$$

Given,

$$FV = \text{Tk. } 1,000$$

$$kd = 0.08$$

$$CIR = 10\%$$

$$MV = \text{Tk. } 1,000$$

$$I = FV \times CIR$$

$$= \text{Tk. } 100$$

Class - 35

14/12/2

Qh-6

Beximco Company issued 5%, 12 years bond of Tk. 1,000 at 5% premium.

The bond is currently selling at 10% discount, $k_d = 12\%$. Determine the intrinsic

value of bond. IF floatation cost is 2%. The bond will be called at 5% discount. Here,

$$FV = \text{Tk. } 1,000$$

$$MV = \text{Tk. } 1,000 - (1,000 \times 10\%)$$

$$= \text{Tk. } 900$$

$$k_d = 12\%$$

$$n = 12$$

$$C/R = 5\%$$

$$F = (1,000 \times 5\%)$$

$$= \text{Tk. } 50$$

$$= \text{Tk. } 50$$

$$V_B < MV, \therefore \text{Call}$$

We know,

$$V_B = \frac{I}{k_d} \times \left\{ 1 - \frac{1}{(1+k_d)^n} \right\} + \frac{MV}{(1+k_d)^n}$$

$$= 50 \times \left\{ \frac{1 - \frac{1}{(1+0.12)^2}}{0.12} \right\} + \frac{500}{(1+0.12)^2}$$

$$= 50 \times 6.1043 + 231.5075$$

$$= \text{Tk. } 788$$

2 Current Yield

$$V_B = \frac{I}{SV}$$

Here,

$I = \text{Coupon interest}$

2 Perpetual Bond

$$V_B = \frac{I}{k_d} = \frac{50}{0.12}$$

$$= \text{Tk. } 750$$

3 Zero-Coupon Bond

$$V_B = \frac{MV}{(1+k_d)^n} = \frac{500}{(1+0.12)^2} = \text{Tk. } 231.507$$

* issued/sold = sale value will change
* matured/redeemed = maturity value

Bond's Yield

1) YTM (Yield to Maturity)

$$YTM = \frac{I + \frac{MV - SV}{n}}{\frac{MV + SV}{2}}$$

Here,

SV =

$$1,000 + (1,000 \times 0.05)$$

$$= \text{Rs. } 1050$$

NSV

$$= 1050 - (1000 \times 0.05)$$

$$= 1025$$

$$= \frac{50 + \frac{1000 - 1050}{2}}{\frac{1000 + 1050}{2}}$$

$$= \frac{50 + \frac{1000 - 1050}{2}}{\frac{1000 + 1050}{2}}$$

$$= 8.2\%$$

$$= 8.2\%$$

Bond

Yield

$$\frac{50}{1025} = \frac{I}{SV} = 4.9\%$$

$$0.049 \times 1025 = 50.225$$

Yield to Maturity
YTM = 8.2%
I = Coupon Interest
SV = Current Value
MV = Maturity Value

Zero-Coupon Bond

$$YTM = \frac{MV - SV}{(1 + YTM)^n}$$

Yield to Maturity (YTM) is the rate of return earned on a bond held to maturity. It is the rate that equates the present value of the bond's cash flows to its current market price.

Q.10

Class - B6

12/12/25

Chapter - 6Ex Current Yield

$$CY = \frac{I}{NSV} = \frac{90}{1030} \times 100$$

$$= 8.73\%$$

Ex Yield to Call

$$YTC = \frac{I + \frac{CP - NSV}{N}}{\frac{CP + NSV}{2}}$$

$$= \frac{90 + \frac{950 - 1030}{8}}{\frac{950 + 1030}{2}}$$

$$= 8.08\%$$

Here,

$$CP = FV - (FV \times DR)$$

$$= 1000 - (1000 \times 5\%)$$

$$= \text{Tk. } 950$$

$$NSV = \text{Tk. } 1030$$

$$I = \text{Tk. } 90$$

Calculation - Present Value of Bond

① Coupon bond [Mention coupon rate and time.]

② Zero coupon bond [missing coupon rate.]

③ Perpetual bond [time is missing]

☐ Coupon bond:

$$i) V_B = I \times \left[\frac{1 - \frac{1}{(1+kd)^n}}{kd} \right] + \frac{MV}{(1+kd)^n}$$

[annual/simple

interest]

$$ii) V_B = \frac{I}{m} \times \left[\frac{1 - \frac{1}{(1+\frac{kd}{m})^{n \times m}}}{\frac{kd}{m}} \right] + \frac{MV}{(1+\frac{kd}{m})^{n \times m}}$$

[compound
interest]

☐ Zero coupon bond:

$$iii) V_B = \frac{MV}{(1+kd)^n}$$

$$iv) V_B = \frac{I}{kd}$$

I = amount of interest

$$I = FV \times CR$$

kd = expected/required rate of return (না থাকলে always 10% ধরে নিবো।)

V_B = value of bond

n = time

m = number of compounding

MV = maturity value

MV - directly mentioned

If not mentioned, face value will be M

If premium rate given,

$$MV = FV + (FV \times PR) \quad \begin{matrix} FV = \text{Face value} \\ PR = \text{Premium Rate} \end{matrix}$$

If discount rate given

$$MV = FV - (FV \times DR) \quad \begin{matrix} FV = \text{Face value} \\ DR = \text{Discount Rate} \end{matrix}$$

to find coupon pay:

$$MV = \frac{FV}{(1+R)^n}$$

$$\frac{FV}{R} = \frac{FV}{R} \times \frac{1}{(1+R)^n}$$

to find interest to be paid

Yield from Bond

~~① YTM~~

① Yield to Maturity, YTM

② Yield to Call, YTC

③ Current Yield, CY

* Formula

$$\textcircled{1} \text{ YTM} = \frac{I + \frac{MV - NSV}{n}}{\frac{MV + NSV}{2}} \times 100$$

[∵ Yearly Interest]

$$\textcircled{2} \text{ YTM} = \frac{\frac{I}{m} + \frac{MV - NSV}{n \times m}}{\frac{MV + NSV}{2}} \times 100$$

[∵ Compound Interest]

$$\textcircled{3} \text{ YTC} = \frac{I + \frac{CP - NSV}{n}}{\frac{CP + NSV}{2}} \times 100$$

$$\textcircled{4} \text{CY} = \frac{I}{\text{NSV}} \times 100$$

Here,

CP = Call Price

$$\text{CP} = \text{FV} + (\text{FV} \times \text{PR}) \quad [\because \text{Premium}]$$

$$\text{CP} = \text{FV} + (\text{FV} \times \text{DR}) \quad [\because \text{Discount}]$$

NSV = Net Sales Value

$$= \text{SV} - \text{FC} \quad [\because \text{FC} = \text{Floatation Cost}]$$

SV = Sales Value

$$= \boxed{\text{SV} - (\text{SV} \times \text{FR})} \quad [\because \text{FR} = \text{Floatation Rate}]$$

→ IF floatation cost is given in %.